First Pandas HW

This homework will take some time.

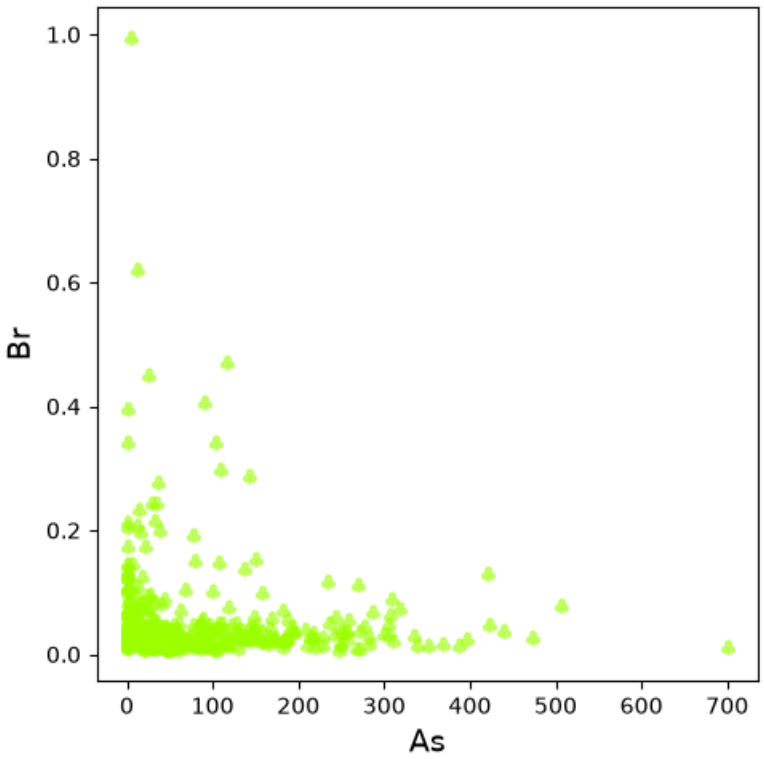
- In a markdown cell write down the AI model you use if needed
- This Homework is following the notes. Use the notes!!!!
- You will see each question and my answers below. Your HW should look very similar but with code and some more epxlanation.
- If you get stuck try the notes. Try and find the cell in the notes where we did something similar.
- If you are truly stuck then come to office hours, talk to a friend, use google or AI for help. If you use help on a question write down what you asked and what you learned
- The last question will have an AI component.
- I am not sure how long this HW will take.
- use the well_data.xlsx file.

AI Model:

Q1

- read in the well_data.xlsx file.
- Make a pdf file that contains a plot of As versus every other parameter.
- Make sure it is labeled nicely and in one pdf file.
- Arsenic should be on the x-axis and the other parameters should be on the y-axis.
- Each parameter should be on one page.
- make sure your axes go in the correct direction. We are not plotting depth anymore
- your graphs should be square.
- Choose a color and symbol you like for your plots
- Only plot the elements. No other parameters.
- Do not plot As vs As
- You will need to hand in your pdf file of results.
- your plot will not show up in the notebook. I am showing mine so you have an example.

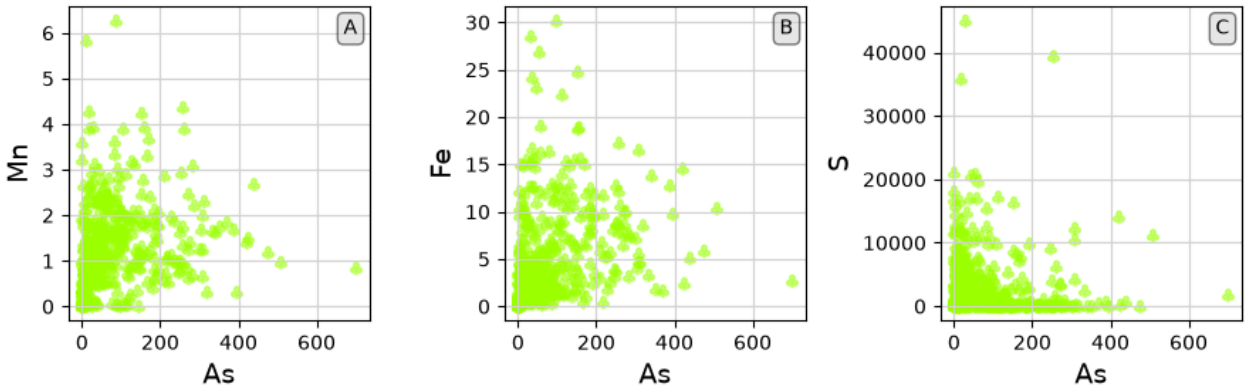
```
Si
P
S
Ca
Fe
Ba
Na
Mg
K
Mn
Sr
F
Cl
SO4
Br
Successfully exported 15 pages to Results/Homework.pdf!
```



Q2

- Make a plot with subplots that is As versus at least 3 other parameters each on their own graph but all on one page.
- You can decide if it is better to share an x or share a y axis. But Arsenic is the independent variable.
- Save this graph to a jpeg and turn that in .
- For this your graphs should be square. Adjust the figure size to make them square.
- They should have letters denoting them like in the packet
- You will need to think about if you want a sharex or a sharey or neither

```
0 Mn
1 Fe
2 S
```



Q3

- Ask AI how your coding is and if there are any improvements you could do.

- when doing this make sure to submit the datafile.
- Answer what AI suggests and then let me know what you try and add a comment to the lines you got from AI.

Bonus Final 10%

- Redo #2 the plot with 3 boxes but color the points by your parameter from the last homework (pump, platform, aquifer, shared).
- You should only need to alter the original scatter call, add a second scatter call, add a legend, and save as a new name.
- Choose the color scheme from your favorite team, company etc and state on the figure or in your notebook what your color scheme represents.

```
0 Mn
1 Fe
2 S
```

